

MAXIMILIAN LARTER

I am a plant ecophysiologicalist and an evolutionary biologist.
My primary research interest is understanding how plants respond and adapt to their environment, especially under climate change. I study functional traits and key physiological processes, and use statistics and models to predict the impacts of climate change on the performance, survival, and future distributions of wild and cultivated species.

RESEARCH EXPERIENCE

- 2025

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Postdoc - modelling urban heat islands and carbon storage
Sylvain Delzon's lab
· Model the impacts of management, development and land use changes on local temperature and carbon storage on the university campus.

INRAE, Bordeaux, France
- 2024
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2021

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Postdoc - Plant hydraulics & trait trade-offs, forest ecology and biogeography
Sylvain Delzon's lab

INRAE, Bordeaux, France
- 2021
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2019

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Postdoc - Herb hydraulics, positive root pressure and drought resistance in Brassicaceae
Frederic Lens's lab

Naturalis, Leiden, The Netherlands
- 2019
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2017

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Postdoc - Evolution of the anthocyanin pathway in Iochrominae
Stacey Smith's lab

University of Colorado, Boulder, USA
- 2016
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2012

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PhD - Evolution of cavitation resistance in conifers
Sylvain Delzon's lab
· Supervisors: Sylvain Delzon and Jean-Christophe Domec

Université de Bordeaux, France

EDUCATION

- 2025
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2022

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Qualification CNU
Sections 66 Physiologie & 67 Biol. des populations et écologie
- 2016
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2012

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PhD in evolutionary, functional and community ecology.
Thesis: Evolution of cavitation resistance in conifers
Université de Bordeaux
· Supervisors: Sylvain Delzon and Jean-Christophe Domec

Bordeaux, France
- 2011

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MSc - Terrestrial Ecosystem Functioning and Modelling
Université de Bordeaux

Bordeaux, France
- 2008

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BSc - Organismal Biology
Université d'Orléans

Orléans, France

SELECTED PUBLICATIONS

- 2025

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M Larter, G Charrier, S Delzon, W Hammond, A Baranger, C Bertrand, N Martin-StPaul, G Kunstler (2025) Weak global trade-off between frost and drought resistance in trees, *New Phytologist*. [pdf](#)
- 2024

●

A Baranger, T Cordonnier, G Charrier, S Delzon, M Larter, N Martin-StPaul, G Kunstler (2024) Living on the edge - physiological tolerance to frost and drought explains range limits of 35 European tree species, *Ecography*. [pdf](#)

CONTACT

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+33 679709275
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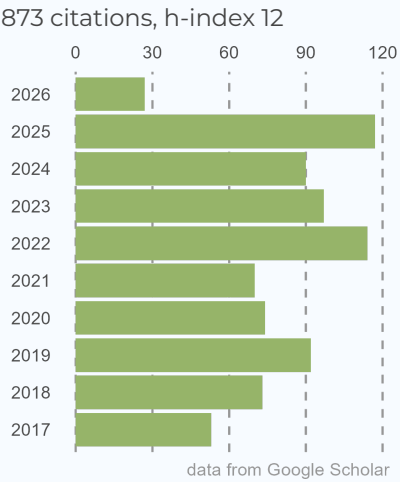
10 rue HG Wells,
33600 Pessac, France
- 📅

August 5th, 1987. Derby (UK)

SKILLS

- Native speaker in English and French
- data analysis, statistics (SAS, R, git), modelling
- QGIS, Inkscape
- Plant hydraulics, physiology, gas exchange
- Molecular biology, phylogenetics

CITATIONS



[full publication list](#)

- 2019 ● M Larter, A Dunbar-Wallis, AE Berardi, SD Smith (2019) Developmental control of convergent floral pigmentation across evolutionary timescales, *Developmental dynamics*, 248 (11), 1091-1100. [pdf](#)
- 2018 ● M Larter, A Dunbar-Wallis, AE Berardi, SD Smith (2018) Convergent evolution at the pathway level: predictable regulatory changes during flower color transitions, *Molecular biology and evolution*, 35 (9), 2159-2169. [pdf](#)
- 2017 ● M Larter, S Pfautsch, JC Domec, S Trueba, N Nagalingum, S Delzon (2017) Aridity drove the evolution of extreme embolism resistance and the radiation of conifer genus *Callitris*, *New Phytologist*, 215 (1), 97-112. [pdf](#)
- 2015 ● M Larter, TJ Brodribb, S Pfautsch, R Burlett, H Cochard, S Delzon (2015) Extreme aridity pushes trees to their physical limits, *Plant Physiology*, 168 (3), 804-807. [pdf](#)
- 2014 ● PS Bouche, M Larter, JC Domec, R Burlett, P Gasson, S Jansen, S Delzon (2014) A broad survey of hydraulic and mechanical safety in the xylem of conifers, *Journal of Experimental Botany*, 65 (15), 4419-4431. [pdf](#)



GRANTS AND FUNDING

- Active participation in grant writing
ANR projects, Calls for Recherche Inter-disciplinaire et Exploratoire and Prématuration-Innovation at Université de Bordeaux, Fondation pour la Recherche et la Biodiversité IDEASHARE & DATASHARE
- 2020 ● Alberta Mennega Stichting fieldwork grant
1,250€ 📍 Leiden (The Netherlands)
- 2014 ● External mobility grant from the COTE Cluster of Excellence
3,000€ 📍 Bordeaux (France)
- Research Exchange Program (Inbound)
AU\$3350 📍 Western Sydney University, NSW (Australia)



CONFERENCES AND PRESENTATIONS

- 2025 ● Poster - "Weak global trade-off between frost and drought resistance in trees"
Xylem International Meeting XIM6 📍 Sevilla (Spain)
- 2022 ● Talk - "Trade off in cold and drought tolerance in trees"
Xylem International Meeting XIM5 📍 Wurzburg (Germany)
- 2017 ● Talk - "Linking changes in gene expression to the macroevolution of flower color in Iochrominae"
Evolution Meeting 📍 Portland, Oregon
- 2015 ● Talk - "The evolution of cavitation resistance in conifers and the case of world-record *Callitris*"
Xylem International Meeting XIM2 📍 Bordeaux (France)



PROFESSIONAL SERVICE

- Guest Lectures (1h)
MSc courses Plant Physiology, Methods in Biodiversity Analysis 📍 Bordeaux U. & Leiden U.
- Supervision of several 6 month student projects
1 BSc and 4 MSc - two leading to publications 📍 Bordeaux U. & Leiden U.
- Regular reviewer for scientific journals
New Phytologist, Tree Physiology, PCE, Evolution, American Journal of Botany, Plos One, BMC Evol. Biol.
- Postdoc and short term contract representative 2021-2022
Conseil d'Unité UMR Biogeco 📍 Bordeaux (France)